OPERATING SYSTEMS **MODULE-13 ASSIGNMENT** **MAXIMUM GRADE: 30**

Q.1) **(2 points)**

State any 2 advantages of using the technique of Segmentation over Paging for memory management.

Q.2) **(4 points)**

Consider a simple segmentation system with a 1K-address maximum segment size that always has access to a byte addressable physical memory. It uses a segment table in which the starting address of any segment regularly happens to be at the real address: (22 + 4096 + Segment No.) Write the binary translation of the logical address 0011100010111010 into a physical address under this memory management scheme. Kindly illustrate your solution with the help of appropriate mathematical steps.

Q.3) **(8 points)**

A process contains 5 distinct virtual pages on disk and is assigned a fixed allocation of 4-page frames in main memory. Assume that the frames are initially empty. The following page trace occurs:

1, 2, 3, 0, 1, 2, 4, 1, 2, 3, 0, 4, 2, 1, 2, 3, 0, 4

A) Show the successive pages residing in the 4 frames using the First-In-First-Out replacement policy.
B) List the number of page faults.
C) Calculate the page fault ratio.
D) List the number of page hits.
E) Calculate the hit ratio.

Q.4) **(8 points)**

Repeat the entire of Q.3 for the Least-Recently-Used replacement policy.

Q.5) **(8 points)**

Repeat the entire of Q.3 for the Optimal replacement policy.